

Manipal Hackathon 2026



Team Name:

Track: Healthcare – Good Health and Well-being

Problem Statement: P10 – From One Empty Shelf to a Regional Shortage

Solution Title: SupplySense – Early Warning System for Medicine Shortages

Team Members:

Solution



SupplySense connects village, block, and district health facilities on one platform to catch medicine shortages early — before an empty shelf becomes a regional crisis.

The Problem



Isolated Tracking

Facilities track stock in isolation — a shortage is invisible to anyone outside that facility until it's already critical



Unconnected Dots

No one connects the dots when multiple facilities run low on the same medicine at the same time



Imbalanced Inventory

Surplus stock sits unused at one facility while a nearby facility stocks out

Our Solution



Shared Dashboard: Multi-level visibility across facility, block, and state levels for stock, consumption, and replenishment



Dual-Alert Algorithm: Flags both local stockout risks and regional shortage patterns



Geographic Map View: Interactive map showing medicine availability with smart redistribution suggestions



Human-in-the-Loop: Every recommendation is human-approved — system informs decisions, doesn't act alone

IMPACT:



Faster response



Fewer stockouts



Less wastage



Better use of existing stock

Technical Implementation



System Structure: Village Health Center → Block Hospital → District Warehouse → State

Two-Way Data Flow: Supply moves downstream (replenishment) • Shortage signals move upstream (alerts)

1. Detection & Analysis Flow



Facility Level Logging

Admins log periodic stock updates (received/dispensed); system computes days-of-stock-remaining per medicine.



Threshold Detection Algorithm

Flags status (Green/Yellow/Red/Critical) based on consumption rate with uncertainty ranges.



Regional Shortage Signal

Cluster trends trigger regional alerts early for systemic disruption across facilities.

2. Action & Response Mechanism



Redistribution Matching

Scans nearby facilities for surplus, ranks matches by distance and urgency, and suggests transfers.



Human-in-the-Loop

District/block managers review and approve every action — system recommends, humans act.



Live Map View

Select any medicine to see real-time availability across all facilities geographically.

ARCHITECTURE:



4-Tier Hierarchy



Dual Signal & Stock Flow



Smart Matching & Map

Feasibility



Tech Stack: React (Frontend) | FastAPI (Backend) | Mapbox (Mapping)

Rule-based detection algorithm (rolling consumption average + threshold logic) — explainable & lightweight, no black-box ML required at pilot stage.

Why It's Buildable Now



Modest Infrastructure Requirements

Core logic (thresholds, trends, geographic matching) runs on basic hardware and functions reliably even with modest internet connectivity.



Immediate Data Validation

Simulated & public health supply data used to thoroughly demonstrate and validate the approach prior to full field deployment.

Rollout Plan



Phase 1: Single District Pilot

Deploy across a handful of key facilities within one supply chain to test end-to-end workflows.



Phase 2: Regional Expansion

Expand coverage to neighboring blocks and districts once core operational workflows are validated.



Phase 3: Model Refinement

Refine detection algorithms using accumulated real usage data to enhance accuracy and lead times.

FEASIBILITY:



Minimal Tech Overhead



Low-Bandwidth Ready



Phased Low-Risk Rollout

Business Strategy



Business Model: Not a consumer product — purpose-built for government health systems and public health programs.

Monetization



State Department Licensing

Subscription & licensing model aligned with existing government procurement channels.



Grant & NGO Pilot Funding

Initial pilot funded via public health donors, transitioning to full state ownership long-term.

Why It Pays For Itself



Lower Emergency Procurement

Prevents costly last-minute stockout orders.



Reduced Medicine Wastage

Redistributes surplus before expiration.



Streamlined Administration

Cuts manual tracking and oversight effort.

Go-to-Market Strategy



Direct Official Engagement

Engage health officials directly with pilot proof.



Small-Scale Validation

Demonstrate impact before state-wide scaling.



System Integration

Partner with existing infrastructure, not replace.

STRATEGY:



B2G Public Health Model



High ROI & Cost Reduction



Public Sector Partnership